



FINAL YEAR DISSERTATION

“Capital Requirements and Credit Supply: The Effect of the RBI's November 2023 Risk-Weight Measures on Consumer Credit and Bank Funding of Indian NBFCs, FY2021–FY2026”

A project report submitted in partial fulfilment of the requirements for the degree of Master of Business Administration (Full-Time) from the Faculty of Management Studies (FMS), University of Delhi.

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CERTIFICATE

Certified that the dissertation entitled “Capital Requirements and Credit Supply: The Effect of the RBI's November 2023 Risk-Weight Measures on Consumer Credit and Bank Funding of Indian NBFCs, FY2021–FY2026”, submitted by Nityam Gupta, MBA (Full-Time), Batch 2025–27, Roll no. FT-25-235, in partial fulfilment of the requirements for the degree of Master of Business Administration of the Faculty of Management Studies, University of Delhi, is a bona fide record of research carried out by the student under my supervision. All sources of data and published work used have been cited. The work has not been submitted for any other degree or diploma.

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DECLARATION

I hereby declare that this dissertation titled “Capital Requirements and Credit Supply: The Effect of the RBI's November 2023 Risk-Weight Measures on Consumer Credit and Bank Funding of Indian NBFCs, FY2021–FY2026” is my original work carried out under the guidance of Dr. Dezy Kumari, Assistant Professor (Finance), Faculty of Management Studies, University of Delhi. It has not been submitted elsewhere for any degree or diploma. All sources of data and published work used have been acknowledged.

Use of AI tools. Artificial intelligence (AI) tools were used for data scraping, that is, extracting figures from the companies' quarterly investor presentations, and for the Python-based data analysis. Every extracted value is recorded with the document and page it came from, so that it can be checked against the original source.

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ABSTRACT

On 16 November 2023 the Reserve Bank of India (RBI) raised capital requirements on two channels of credit at once: the risk weight on unsecured consumer credit (personal loans and credit cards) and the risk weight on banks' lending to non-banking financial companies (NBFCs), with housing finance companies (HFCs) exempt. This dissertation asks whether these targeted increases in capital requirements reduced the supply of credit through each channel, and how listed NBFCs adjusted. Using monthly RBI data on the sectoral deployment of bank credit (January 2019 to July 2026), I compare credit categories hit by the higher risk weights with exempt categories before and after the policy. Growth of bank credit to the targeted consumer categories fell by 0.66 percentage points per month relative to exempt household credit (about 7.9 percentage points a year; $p = 0.002$), and bank credit to NBFCs (excluding HFCs) slowed by 1.00 percentage points per month relative to banks' total non-food credit (about 11.9 percentage points a year; $p = 0.049$). Placebo tests using a fictitious November 2022 policy date show no effect. The bank-funding result is robust to controlling for seasonality and to alternative inference methods; the consumer-credit result is consistent in direction and size but its significance depends on correcting for serial correlation in the monthly data (p between about 0.06 and 0.18 under some alternative methods). To trace how NBFCs adjusted, I hand-collected quarterly data from 382 investor presentations of 16 listed NBFCs and HFCs. Two lenders disclose that the rule cut their capital ratios by roughly 3 to 4 percentage points, and one reports raising lending rates. Firm-level tests show bank-dependent NBFCs facing somewhat higher funding costs and non-HFC NBFCs drifting away from bank borrowing relative to exempt HFCs, but these differences are not statistically significant in a sample of 16 firms. The evidence indicates that the RBI's risk-weight measures worked mainly through the banking system: banks cut back targeted consumer credit and lending to NBFCs, while listed NBFCs absorbed the shock through capital, pricing and gradual funding substitution rather than by shrinking their own lending.

Keywords: risk weights, capital requirements, credit supply, NBFCs, consumer credit, macroprudential policy, difference-in-differences, India

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EXECUTIVE SUMMARY

Context. In November 2023 the Reserve Bank of India raised the capital that lenders must hold against unsecured consumer loans and credit cards, and the capital that banks must hold against their loans to NBFCs (housing finance companies were exempt). The measures were aimed at the two fastest-growing parts of the credit system. The bank-to-NBFC measure was withdrawn from April 2025; the consumer measure was not.

Objective. To measure whether the higher risk weights reduced credit through each targeted channel, and to document how listed NBFCs adjusted.

Data. Monthly RBI data on sectoral deployment of bank credit (2019-01-18 to 2026-07-31), and a firm-level dataset hand-collected from 382 quarterly investor presentations of 16 listed NBFCs and HFCs (cost of funds, bank share of borrowings, segment assets and company statements).

Method. Difference-in-differences: credit categories that faced the higher risk weight are compared with closely related exempt categories before and after the policy, with placebo tests on a fictitious policy date, robustness checks and two-way fixed-effects firm panels with wild cluster bootstrap inference.

Key findings.

- Model 1 (H1): bank credit to cards and other personal loans grew about 7.9 percentage points a year more slowly than exempt household credit after the policy (Newey–West $p = 0.002$; panel $p = 0.008$). The direction is the same under every specification, but significance is weaker under methods that treat serial correlation differently (Section 8.5).
- Model 2 (H2): bank credit to NBFCs (excluding HFCs) grew about 11.9 percentage points a year more slowly than total bank credit ($p = 0.049$), and the result holds when calendar-month seasonality is controlled for. It recovered strongly after the April 2025 rollback.
- Placebo tests on November 2022 show no effect. The implied shortfalls by January 2025 are roughly ₹1.9 lakh crore of targeted consumer credit and ₹2.1 lakh crore of bank funding to NBFCs.
- Listed NBFCs absorbed the shock through capital (capital ratios cut by about 3–4 percentage points at the two lenders that disclosed it), pricing (rates raised 20–30 bps at one lender), somewhat higher funding costs and a gradual move away from bank borrowing; these firm-level effects are in the expected direction but not statistically significant with 16 firms.

Implications. Targeted risk weights are an effective and reversible tool for slowing specific segments of bank credit. Their effect on NBFCs depends on how easily NBFCs can replace bank funding, so diversified funding is a key defence for NBFC managers.

RATIONALE OF THE STUDY

NBFCs now supply a large share of India's retail credit and depend heavily on banks for funding. A regulatory change that raises the cost of both their consumer lending and their bank borrowing therefore matters to the regulator, to banks, to NBFC managers and to investors. The study is worthwhile for four reasons.

1. **Policy relevance.** The November 2023 measures were one of the RBI's most significant macroprudential interventions since the IL&FS crisis, and were partly reversed in 2025. Whether they worked, and through which channel, is directly relevant to how such tools are used in future.
2. **A clean natural experiment.** The circular changed capital requirements for clearly defined categories on a known date, left comparable categories untouched and was later partly withdrawn. This allows the effect of the rule to be separated from the interest-rate cycle and other common shocks.
3. **Managerial relevance.** For NBFC managers the episode shows how exposure to a single funding source translates into regulatory risk, and how capital, pricing and funding mix can be used to absorb such a shock.
4. **Gap in evidence.** Existing discussion of the measures is largely descriptive. There is little formal evidence that separates the consumer-credit and bank-funding channels or that uses NBFCs' own disclosures to trace their response.

1. INTRODUCTION

1.1 Background and motivation

Non-banking financial companies (NBFCs) have become one of the main channels of retail credit in India. Between 2021 and 2023, unsecured consumer lending — personal loans and credit cards — grew far faster than overall credit, and NBFCs both lent directly to households and borrowed heavily from banks to fund that lending. The Reserve Bank of India (RBI) viewed this combination as a build-up of risk: rapid unsecured lending on the asset side, and growing interdependence between banks and NBFCs on the funding side.

On 16 November 2023 the RBI responded with a single circular that raised capital requirements on both channels. First, it increased the risk weight on consumer credit — excluding housing, education, vehicle, gold-backed and microfinance loans — from 100 per cent to 125 per cent for banks and NBFCs, and raised risk weights on credit card receivables. Second, it increased by 25 percentage points the risk weight that banks attach to their loans to most NBFCs, while exempting housing finance companies (HFCs). Because a higher risk weight requires a lender to hold more capital against the same loan, both measures made the targeted credit more expensive to supply. In February 2025 the RBI partially reversed course, withdrawing the higher risk weight on bank loans to NBFCs from 1 April 2025, but leaving the consumer-credit measures in place.

This policy is a useful natural experiment. It changed capital requirements for clearly defined categories of credit at a known date, left closely comparable categories untouched, and was later partly reversed. That design makes it possible to separate the effect of the rule from general credit conditions such as the interest-rate cycle.

1.2 Research problem

Market commentary since 2023 suggests that unsecured lending slowed after the RBI's measures, but it is not clear how much of the slowdown was caused by the rule itself, which channel it worked through, and how NBFCs responded. The higher risk weights could have reduced NBFCs' own lending, raised their borrowing costs, pushed them away from bank funding, or been absorbed through lower capital buffers and higher lending rates. This study measures the effect on each channel and documents how listed NBFCs adjusted.

1.3 Objectives and hypotheses

Following the guide's direction to focus on one or two hypotheses, the study tests the two levers of the November 2023 circular:

- **H1 (consumer-credit channel):** Higher risk weights on consumer credit reduced the growth of credit to the targeted consumer categories (credit cards and other personal loans) relative to exempt household credit categories.
- **H2 (bank-funding channel):** Higher risk weights on bank exposures to NBFCs reduced the growth of bank credit to NBFCs (excluding the exempt HFCs) relative to banks' overall lending.

These hypotheses correspond to H1 and H2 of the original synopsis, re-specified so that each one is tested on the group that the regulation actually targeted. The synopsis versions — firm-level effects of NBFCs' pre-policy unsecured share and bank dependence — are tested as supporting evidence using a hand-collected firm-level dataset (Chapter 8).

1.4 Contribution

The study makes three contributions. First, it provides a clean before-and-after comparison of targeted and exempt credit categories using official RBI data, with placebo tests. Second, it assembles a new firm-level dataset from 382 quarterly investor presentations of 16 listed NBFCs and HFCs, including cost of funds, bank share of borrowings and company disclosures of the policy's capital impact. Third, it shows that an initial firm-level dataset obtained for the project was materially inaccurate, and documents how it was verified and replaced — a practical lesson for empirical work on Indian NBFCs.

1.5 Structure

Chapter 2 describes the regulatory background and Chapter 3 reviews the literature. Chapter 4 describes the data and how they were verified, including descriptive statistics. Chapter 5 sets out the econometric methodology and Chapter 6 the theoretical framework, variables and formal hypotheses. Chapters 7 and 8 present the empirical findings for H1 and H2, together with the firm-level evidence, model diagnostics and the sensitivity of inference. Chapter 9 discusses the findings and limitations, and Chapter 10 concludes.

2. REGULATORY BACKGROUND

2.1 Risk weights and capital requirements

Under the Basel framework applied in India, banks and NBFCs must hold regulatory capital equal to a minimum percentage of their risk-weighted assets (RWA). Each exposure is multiplied by a risk weight; a 125 per cent risk weight means that every ₹100 lent counts as ₹125 of RWA. Raising the risk weight therefore increases the capital a lender must hold against the same loan or, equivalently, lowers its capital-to-risk-weighted-assets ratio (CRAR) if capital does not change. For an NBFC that must keep CRAR above 15 per cent, or a bank that must meet its own capital requirements, a higher risk weight raises the effective cost of supplying that type of credit.

2.2 The November 2023 circular

The RBI's circular of 16 November 2023 (“Regulatory measures towards consumer credit and bank credit to NBFCs”) contained the measures summarised in Table 2.1.

Table 2.1: Main risk-weight changes, November 2023 and February 2025

Measure	Change	Exemptions / notes
Consumer credit (banks and NBFCs)	Risk weight 100% → 125%	Excludes housing, education, vehicle, gold-backed and microfinance/SHG loans
Credit card receivables	Banks 125% → 150%; NBFCs 100% → 125%	—
Bank exposures to NBFCs	+25 percentage points where the rating-based risk weight was below 100%	Excludes HFCs and loans eligible for priority-sector classification
February 2025 circular	Additional risk weight on bank exposures to NBFCs withdrawn from 1 April 2025	Consumer-credit risk weights unchanged; microfinance treatment eased

Source: RBI circulars of 16 November 2023 and 25 February 2025 (see Bibliography).

Two features of the circular matter for the empirical design. The consumer-credit measure targeted unsecured personal loans and cards but left housing, vehicle and education loans untouched, so exempt household categories provide a natural comparison group for H1. The bank-to-NBFC measure applied to NBFCs but not to HFCs, so bank credit to HFCs and to the banking system as a whole provide comparison groups for H2. The February 2025 rollback of the NBFC measure, with no rollback of the consumer measure, provides a further check on the targeted nature of the effects.

2.3 Macroeconomic setting

The policy came after the RBI had raised the repo rate by 250 basis points between May 2022 and February 2023, to 6.5 per cent, where it stayed until February 2025. Funding costs across the financial system were therefore already rising during 2022–23 as loans repriced. This is the main

reason why a comparison with untargeted credit categories, rather than a simple before-and-after comparison, is necessary.

2.4 The NBFC sector before the policy

NBFCs are regulated by the RBI but, unlike banks, cannot accept demand deposits and do not have access to the RBI's liquidity facilities in the same way. They fund themselves through bank loans, bonds (non-convertible debentures), commercial paper, securitisation and, for some, public deposits. The default of IL&FS in 2018 showed how quickly funding can dry up for NBFCs that depend on short-term market borrowing, and banks subsequently became an even more important source of NBFC funding. In October 2021 the RBI introduced a Scale-Based Regulation framework, placing NBFCs in base, middle, upper and top layers with progressively stricter requirements; most of the firms in this study fall in the middle or upper layers.

By October 2023 the two channels targeted by the RBI were large. Outstanding bank credit was ₹2.4 lakh crore to credit cards and ₹13.3 lakh crore in other personal loans, and bank credit to NBFCs excluding HFCs stood at ₹11.2 lakh crore — about 7.2 per cent of banks' total non-food credit. All three had been growing at more than 20 per cent a year (Table 4.1), well above overall credit growth. The circular was therefore aimed at the fastest-growing parts of the credit system, and at the link between banks and NBFCs that had become central to NBFC funding after 2018.

3. LITERATURE REVIEW

3.1 Capital requirements and credit supply

A large literature shows that capital requirements affect the supply of bank credit. Kashyap and Stein (2004) explain how risk-sensitive capital rules can amplify credit cycles. Aiyar, Calomiris and Wieladek (2014) use changes in bank-specific capital requirements in the United Kingdom to show that higher requirements reduce lending, while part of the gap is filled by foreign branches outside the regulator's reach. Jiménez, Ongena, Peydró and Saurina (2017) find that Spain's dynamic provisioning — a capital buffer — dampened credit cycles, with effects concentrated in more constrained banks. Behn, Haselmann and Wachtel (2016) show that when model-based risk weights rose during the 2008 crisis, German banks cut lending more for loans whose risk weights increased. Together these studies suggest that raising risk weights on a category of loans should reduce credit supply to that category, which is the logic behind H1.

3.2 Bank funding and shadow banks

The bank lending channel literature (Bernanke and Blinder, 1988; Khwaja and Mian, 2008) shows that shocks to banks' ability or willingness to lend are passed on to borrowers who depend on bank credit. Khwaja and Mian (2008) use Pakistani loan-level data to show that firms borrowing from affected banks could not fully replace the lost funding. Irani, Iyer, Meisenzahl and Peydró (2021) show that tighter capital regulation of banks shifts lending activity toward non-bank lenders. In India, Acharya, Khandwala and Öncü (2013) document the close funding links between banks and NBFCs. Raising the risk weight on bank loans to NBFCs is a direct shock to this channel; H2 tests whether it reduced bank credit to NBFCs.

3.3 Macroprudential policy and targeted risk weights

Higher risk weights on specific loan categories are a macroprudential tool: instead of tightening credit across the economy through interest rates, the regulator raises the cost of supplying the type of credit it considers risky. The studies above suggest three predictions for such a measure. First, credit growth in the targeted category should slow relative to similar untargeted categories. Second, lenders may respond not only by lending less but by raising prices, raising capital or shifting toward exempt lending. Third, part of the targeted activity may move to lenders or funding sources outside the measure's reach (Aiyar, Calomiris and Wieladek, 2014; Irani et al., 2021). The November 2023 circular is unusual because it targeted both a type of borrower (unsecured consumers) and a type of lender-funding link (banks to NBFCs) at the same time, allowing both channels to be examined within one policy.

3.4 Methodological literature

Policy evaluations of this kind typically compare affected and unaffected groups before and after the change (difference-in-differences). Two inference issues are relevant here. With aggregate monthly series, errors are serially correlated, so standard errors robust to autocorrelation (Newey and West, 1987) or to cross-sectional and serial dependence in panels (Driscoll and Kraay, 1998) are needed. With a small number of firms, conventional cluster-robust standard errors over-reject; the wild cluster bootstrap-t (Cameron, Gelbach and Miller, 2008), with six-point weights for very few clusters (Webb, 2023), is the recommended remedy. Callaway, Goodman-Bacon and Sant'Anna (2024) discuss the interpretation of designs with continuous treatment intensity, relevant to the firm-level exposure tests.

3.5 Research gap

Existing evidence on the November 2023 measures is largely descriptive (RBI Financial Stability Reports and market commentary). There is little formal evidence that separates the consumer-credit and bank-funding channels, uses exempt categories as controls, or documents how listed NBFCs adjusted using their own disclosures. This study addresses that gap.

4. DATA

4.1 Aggregate data: RBI sectoral deployment of bank credit

The main dataset is the RBI's monthly table “Deployment of Gross Bank Credit by Major Sectors” (Database on Indian Economy, Table 15), covering 91 monthly observations from 2019-01-18 to 2026-07-31. It reports outstanding bank credit (₹ crore) to credit cards, other personal loans, housing, vehicle loans, education, loans against gold jewellery, NBFCs, HFCs and total non-food credit. Bank credit to NBFCs excluding HFCs is computed as NBFC credit minus HFC credit. Growth is measured as the monthly change in the natural logarithm of outstanding credit ($\times 100$, approximately per cent per month).

One adjustment is required. On 1 July 2023 HDFC Ltd, then India's largest HFC, merged into HDFC Bank. Its home loans moved onto a bank balance sheet and its bank borrowings disappeared, producing a one-off jump in bank housing credit and a fall in bank credit to HFCs and NBFCs in July 2023. The RBI notes that data from 28 July 2023 include the impact of the merger. The July 2023 growth observation is therefore excluded for housing, HFC, NBFC and non-food credit. Because all tests use month-on-month growth, the level shift does not affect other months.

Table 4.1: Annualised growth of bank credit by category (per cent per year)

Category	Pre-policy (Apr-21 to Oct-23)	Post-policy (Dec-23 to Mar-25)	After rollback (Apr-25 to Jul-26)
Credit cards (targeted, H1)	23.3	11.3	3.5
Other personal loans (targeted, H1)	21.6	10.0	11.3
Housing (exempt)	13.9	11.2	9.8
Vehicle loans (exempt)	14.6	9.5	15.4
Education (exempt)	13.5	15.3	11.8
Gold loans (exempt)	6.8	62.4	73.9
NBFCs ex-HFC (targeted, H2)	23.5	9.1	21.4
HFCs (exempt)	7.1	4.8	12.6
Total non-food credit (benchmark, H2)	13.2	11.4	13.4

Average monthly log growth $\times 12$. July 2023 excluded for merger-affected series. Source: RBI DBIE Table 15; author's calculations.

Table 4.1 already shows the pattern the hypotheses predict. Bank credit to credit cards grew 23.3 per cent a year before the policy and 11.3 per cent after; other personal loans slowed from 21.6 to 10.0 per cent. Exempt categories slowed much less (housing 13.9 $\rightarrow$ 11.2, vehicle 14.6 $\rightarrow$ 9.5) or accelerated (education). Bank credit to NBFCs excluding HFCs slowed from 23.5 to 9.1 per cent, while total non-food credit only eased from 13.2 to 11.4 per cent. After the April 2025 rollback, bank credit to NBFCs recovered to 21.4 per cent, whereas card credit — whose higher risk weight was not reversed — slowed further to 3.5 per cent. Bank gold loans, which were exempt, grew very

rapidly after the policy (62.4 per cent a year), consistent with lenders shifting toward secured credit.

4.2 Firm-level data: hand-collected from investor presentations

The firm-level sample consists of 16 listed NBFCs and HFCs: six HFCs (AAVAS Financiers, Can Fin Homes, Home First Finance, LIC Housing Finance, PNB Housing Finance and Repco Home Finance) and ten NBFCs (Arman Financial Services, Bajaj Finance, Cholamandalam Investment & Finance, CreditAccess Grameen, IIFL Finance, Mahindra & Mahindra Financial Services, Muthoot Finance, Poonawalla Fincorp, SBI Cards and Payment Services, and Shriram Finance). For each company, all quarterly investor presentations from Q1 FY2021 to Q4 FY2026 were collected — 382 documents in total. From these, the following were hand-collected, each value recorded with the presentation and page it came from:

- **Cost of funds** (366 company-quarters, all 16 firms): the company's reported cost of borrowing; where not reported, annualised finance cost divided by average borrowings from the same presentation.
- **Bank share of borrowings** (286 company-quarters, 15 firms): the share of borrowings from banks shown in each company's borrowing-mix slide (Poonawalla Fincorp does not disclose its mix).
- **Pre-policy bank dependence**: the bank share of borrowings at 30 September 2023, verified against each Q2 FY2024 presentation.
- **Bajaj Finance segment-wise assets under management** (24 quarters), separating consumer segments affected by the rule from exempt segments.
- **Company statements** quantifying the effect of the RBI's measures on capital ratios and pricing.

4.3 Data verification

The project began with a pre-compiled firm-level dataset. Checking it against the companies' own filings showed that several key fields were not reported data. For five lenders, “unsecured” assets were a fixed percentage of total assets in every quarter; for Shriram Finance, the series mixed an assumed 9 per cent share with two quarters of actual personal-loan figures, producing artificial spikes. Bank borrowing shares differed materially from the filings for several firms (for example Arman Financial 71 per cent in the dataset versus 32 per cent in its filing; Muthoot Finance 41 versus 65 per cent). Headline figures such as total assets and gross NPA ratios were generally accurate.

All firm-level variables used in this dissertation were therefore re-collected from primary sources. An early test using the unverified bank shares appeared to show a significant effect on borrowing costs; with verified shares the effect disappeared. This illustrates why primary-source verification was essential, and the original dataset is not used in any result reported here.

Table 4.2: Firm sample and pre-policy bank dependence (30 September 2023)

Company	Type	Bank share in original dataset (%)	Bank share verified from filing (%)
AAVAS Financiers	HFC	44.5	49.6
Arman Financial Services	NBFC	71.0	32.3
Bajaj Finance	NBFC	23.0	23.0
Can Fin Homes	HFC	54.0	57.0
Cholamandalam Inv. & Fin.	NBFC	53.5	60.0
CreditAccess Grameen	NBFC (MFI)	58.0	52.9
Home First Finance	HFC	57.5	54.0
IIFL Finance	NBFC	56.0	55.3
LIC Housing Finance	HFC	24.5	33.0
M&M Financial Services	NBFC	56.5	57.2
Muthoot Finance	NBFC (gold)	41.0	65.4
PNB Housing Finance	HFC	40.2	40.2
Poonawalla Fincorp	NBFC	64.0	not disclosed
Repco Home Finance	HFC	74.0	75.5
SBI Cards & Payment	NBFC (cards)	68.5	77.0
Shriram Finance	NBFC	30.5	25.6

Definitions follow each company's borrowing-mix slide (e.g. some include financial institutions). Source: Q2 FY2024 investor presentations; Shriram from its Q2 FY2025 presentation.

Table 4.3: Descriptive statistics of hand-collected firm-level variables

Variable	Group	Period	N	Mean	SD	Min	Max
Cost of funds (%)	Non-HFC NBFCs	Pre (to Q2 FY24)	124	8.13	1.73	4.90	14.87
Cost of funds (%)	Non-HFC NBFCs	Post (Q3 FY24 on)	98	8.57	1.42	6.40	12.70
Cost of funds (%)	HFCs	Pre (to Q2 FY24)	84	7.30	0.65	5.56	8.50
Cost of funds (%)	HFCs	Post (Q3 FY24 on)	60	7.93	0.45	7.07	8.90
Bank share of borrowings (%)	Non-HFC NBFCs	Pre (to Q2 FY24)	99	48.46	14.95	24.00	77.00
Bank share of borrowings (%)	Non-HFC NBFCs	Post (Q3 FY24 on)	83	49.49	19.28	17.37	86.00
Bank share of borrowings (%)	HFCs	Pre (to Q2 FY24)	59	47.52	16.59	20.00	75.50

Variable	Group	Period	N	Mean	SD	Min	Max
Bank share of borrowings (%)	HFCs	Post (Q3 FY24 on)	45	54.64	17.88	31.00	85.50

Company-quarters, Q1 FY2021 – Q4 FY2026. Pre = up to Q2 FY2024 (September 2023); Post = Q3 FY2024 onward.

Table 4.3 summarises the firm-level data. Average cost of funds rose for both groups after the policy — from 8.13 to 8.57 per cent for non-HFC NBFCs and from 7.30 to 7.93 per cent for HFCs — reflecting the general rise in interest rates. The bank share of borrowings, by contrast, rose by about 7.1 percentage points for HFCs but only 1.0 for non-HFC NBFCs. These raw differences motivate the formal tests in Chapter 8.

4.4 Descriptive statistics and correlations of the RBI series

Table 4.4: Descriptive statistics of monthly bank credit growth (per cent per month)

Series	Period	N	Mean	SD	Min	Max
Credit cards	Pre-policy	31	1.94	3.07	-7.58	7.98
Credit cards	Post-policy	14	1.26	1.55	-1.74	3.46
Other personal loans	Pre-policy	31	1.80	1.19	-0.80	4.04
Other personal loans	Post-policy	14	0.84	0.87	-0.78	2.61
Housing	Pre-policy	30	1.16	0.86	-0.43	3.49
Housing	Post-policy	14	0.93	0.44	0.41	1.77
Vehicle loans	Pre-policy	31	1.22	1.36	-1.61	4.14
Vehicle loans	Post-policy	14	0.82	0.75	-0.45	2.64
Education	Pre-policy	31	1.13	1.22	-0.86	4.22
Education	Post-policy	14	1.38	0.81	-0.21	2.56
NBFCs ex-HFC	Pre-policy	30	1.96	3.14	-4.38	9.28
NBFCs ex-HFC	Post-policy	14	0.76	1.93	-2.88	3.81
HFCs	Pre-policy	30	0.59	2.43	-3.45	7.22
HFCs	Post-policy	14	0.51	2.47	-2.59	7.39
Non-food credit	Pre-policy	30	1.10	1.11	-0.93	3.67
Non-food credit	Post-policy	14	0.86	0.77	-0.55	2.38

Monthly log growth $\times 100$. Pre-policy: April 2021 – October 2023; post-policy: December 2023 – January 2025. July 2023 excluded for merger-affected series. Source: RBI DBIE Table 15.

Table 4.4 shows that monthly growth of the targeted series is volatile: the standard deviation of monthly credit-card growth is larger than its mean. This volatility, much of it seasonal (festive-season card spending, quarter-end lending to NBFCs), is the main reason why the regression models in Chapters 7 and 8 explain only a small share of month-to-month variation even when the post-policy shift is large.

Table 4.5: Correlation matrix of monthly bank credit growth, April 2021 – January 2025

	Cards	Oth. PL	Housing	Vehicle	Educ.	NBFC ex-HFC	HFCs	Non-food
Cards	1.00	0.50	0.26	0.36	0.19	0.33	0.37	0.18
Oth. PL	0.50	1.00	0.51	0.19	0.26	0.53	0.49	0.53
Housing	0.26	0.51	1.00	0.31	0.25	0.44	0.27	0.41
Vehicle	0.36	0.19	0.31	1.00	0.27	0.38	0.13	0.11
Educ.	0.19	0.26	0.25	0.27	1.00	0.23	0.14	0.27
NBFC ex-HFC	0.33	0.53	0.44	0.38	0.23	1.00	0.28	0.66
HFCs	0.37	0.49	0.27	0.13	0.14	0.28	1.00	0.14
Non-food	0.18	0.53	0.41	0.11	0.27	0.66	0.14	1.00

Pearson correlations of monthly log growth rates.

Correlations between categories are low to moderate, so each series carries largely independent information. The treated series are not strongly correlated with the control series, which is why the control groups are chosen on economic grounds (similar borrowers, exempt from the rule) rather than on statistical co-movement, and why the tests compare average growth before and after the policy rather than relying on month-to-month co-movement.

5. ECONOMETRIC METHODOLOGY

5.1 Research design

The study uses a difference-in-differences design. For each hypothesis, credit that the regulation targeted (the treatment group) is compared with closely related credit that was exempt (the control group), before and after the policy. Any change in the growth gap between the two groups after November 2023 is attributed to the policy, under the assumption that, without the policy, the gap would have stayed at its pre-policy level. Interest-rate changes, economic growth and other common shocks affect both groups and cancel out in the comparison.

The main sample runs from April 2021 to January 2025, so that the post-policy period ends before the February 2025 rollback was announced. The first post-policy observation is the end of November 2023.

5.2 Aggregate tests

Interrupted time series on the growth gap. For each month t , the growth gap is the average monthly growth of the treated categories minus that of the control categories. The gap is regressed on a constant and a post-policy indicator:

$$Gap(t) = \alpha + \delta \cdot Post(t) + \varepsilon(t)$$

α is the average pre-policy gap and δ is the change in the gap after the policy — the difference-in-differences estimate. Standard errors are Newey–West with six lags to allow for autocorrelation and seasonality. Because the sample is short, Section 8.5 checks the results against classical, heteroskedasticity-robust and bootstrap standard errors and against a specification with calendar-month dummies.

Panel difference-in-differences. As a second specification for H1, each category is treated as a separate series in a panel, with category fixed effects, month fixed effects and a Treated $\times$ Post interaction, estimated with the linearmodels PanelOLS estimator and Driscoll–Kraay standard errors (bandwidth six):

$$g(i,t) = \mu(i) + \lambda(t) + \delta \cdot Treated(i) \times Post(t) + \varepsilon(i,t)$$

Placebo tests. Both tests are repeated using only pre-policy data (April 2021 to October 2023) with a fictitious policy date of November 2022. A significant placebo effect would indicate that the groups were already diverging before the policy.

5.3 Hypothesis definitions

Table 5.1: Treatment and control groups

Hypothesis	Treated (higher risk weight)	Control (exempt)	Expected sign of δ
H1 consumer credit	Credit cards; other personal loans	Housing; vehicle loans; education	Negative
H2 bank funding	Bank credit to NBFCs excluding HFCs	Total non-food bank credit	Negative
H2 alternatives	Bank credit to NBFCs ex-HFC / all NBFCs	Bank credit to HFCs / total non-food credit	Negative

5.4 Identifying assumptions and threats

The key assumption is that, without the policy, the growth gap between treated and control credit would have stayed at its pre-policy average. Four threats are considered. (i) Pre-existing trends: if targeted credit was already slowing relative to controls, the estimate would be biased; the placebo tests and Figures 7.2 and 8.1 check this. (ii) Other shocks at the same time: the interest-rate cycle and economic growth affect treated and control credit alike and are differenced out; the repo rate was unchanged at 6.5 per cent throughout the main post-policy window. (iii) Spillovers: if banks redirected lending from targeted to exempt categories, the control group would grow faster and the estimate would overstate the absolute fall in targeted credit, though it would still measure the policy's effect on the relative allocation of credit. (iv) The HDFC merger in July 2023 changes the level of some series; using month-on-month growth and excluding the merger month removes its effect, and a robustness check drops the housing series entirely.

5.5 Firm-level tests

Firm-level panels are estimated with two-way fixed effects (company and quarter) using PanelOLS. With 16 firms, inference uses the wild cluster bootstrap-t clustered by company (Webb six-point weights, 19,999 draws, null imposed). Three specifications are estimated: (i) cost of funds on $\text{Post} \times \text{verified pre-policy bank share}$; (ii) bank share of borrowings on $\text{Post} \times \text{non-HFC indicator}$, using the regulatory exemption of HFCs as the source of variation; and (iii) for Bajaj Finance, segment-level asset growth on $\text{Post} \times \text{affected-segment indicator}$, with segment and quarter fixed effects.

6. THEORETICAL FRAMEWORK

6.1 Conceptual framework

A risk weight determines how much capital a lender must hold against each rupee of a loan. When the risk weight on a category of loans rises, the lender either has to hold more capital (which is costly) or lend less in that category. The November 2023 circular raised risk weights on two links in the credit chain at once. Table 6.1 traces each link from the regulatory change to the variable that is observed in the data and the hypothesis that tests it.

Table 6.1: Transmission channels, observable variables and hypotheses

Channel	Mechanism	Observable variable	Model
Consumer credit	Higher capital per rupee of unsecured consumer loans → banks slow or reprice such lending	Monthly growth of bank credit to cards and other personal loans relative to exempt household credit	Model 1 (H1)
Bank funding of NBFCs	Higher capital per rupee lent to NBFCs → banks slow lending to NBFCs (HFCs exempt)	Monthly growth of bank credit to NBFCs ex-HFC relative to total non-food credit	Model 2 (H2)
NBFC funding cost	Bank-dependent NBFCs face dearer or scarcer bank loans	Quarterly cost of funds of each NBFC	Model 3
NBFC funding mix	Non-exempt NBFCs substitute bonds, CP and other funding for bank loans	Bank share of each NBFC's borrowings	Model 4
NBFC lending mix	NBFCs slow affected consumer segments relative to exempt segments	Segment-wise AUM growth of Bajaj Finance	Model 5

6.2 Variables

Table 6.2: Definition of variables

Variable	Type	Definition	Source
Gap1(t)	Dependent (Model 1)	Average monthly log growth ($\times 100$) of bank credit to credit cards and other personal loans minus that of housing, vehicle and education loans	RBI DBIE Table 15
Gap2(t)	Dependent (Model 2)	Monthly log growth ($\times 100$) of bank credit to NBFCs excluding HFCs minus that of total non-food bank credit	RBI DBIE Table 15
g(i,t)	Dependent (panel)	Monthly log growth ($\times 100$) of bank credit to category i	RBI DBIE Table 15
COF(f,q)	Dependent (Model 3)	Cost of funds of firm f in quarter q, per cent	Investor presentations
BankShare(f,q)	Dependent (Model 4)	Bank borrowings as a share of total borrowings, per cent	Investor presentations
AUMgrowth(s,q)	Dependent (Model 5)	Quarterly log growth of Bajaj Finance segment AUM	Investor presentations
Post(t)	Independent	1 from end-November 2023 (Q3 FY2024 for quarterly data), 0 before	RBI circular

Variable	Type	Definition	Source
Treated(i) × Post(t)	Independent	1 for targeted categories / firms / segments after the policy	RBI circular
BankShare2023(f) × Post(q)	Independent (Model 3)	Verified bank share at 30 September 2023 interacted with Post	Q2 FY2024 presentations
NonHFC(f) × Post(q)	Independent (Model 4)	1 for NBFCs other than HFCs after the policy	RBI circular
Fixed effects	Control	Category / firm / segment and month / quarter effects; month-of-year dummies in robustness tests	—

6.3 Model 1: consumer-credit channel (H1)

$$Gap1(t) = \alpha_1 + \delta_1 \cdot Post(t) + \varepsilon(t)$$

Null hypothesis H0: $\delta_1 = 0$ — the higher risk weight did not change the growth of targeted consumer credit relative to exempt household credit.

Alternative hypothesis H1: $\delta_1 < 0$ — growth of targeted consumer credit fell relative to exempt household credit after the policy.

6.4 Model 2: bank-funding channel (H2)

$$Gap2(t) = \alpha_2 + \delta_2 \cdot Post(t) + \varepsilon(t)$$

Null hypothesis H0: $\delta_2 = 0$ — the higher risk weight on bank loans to NBFCs did not change the growth of bank credit to NBFCs relative to total bank credit.

Alternative hypothesis H1: $\delta_2 < 0$ — growth of bank credit to NBFCs (excluding HFCs) fell relative to total bank credit after the policy.

6.5 Supporting firm-level models (Models 3–5)

$$Model\ 3: COF(f,q) = \mu(f) + \lambda(q) + \beta_3 \cdot BankShare2023(f) \times Post(q) + \varepsilon(f,q)$$

$$Model\ 4: BankShare(f,q) = \mu(f) + \lambda(q) + \beta_4 \cdot NonHFC(f) \times Post(q) + \varepsilon(f,q)$$

$$Model\ 5: AUMgrowth(s,q) = \mu(s) + \lambda(q) + \beta_5 \cdot Affected(s) \times Post(q) + \varepsilon(s,q)$$

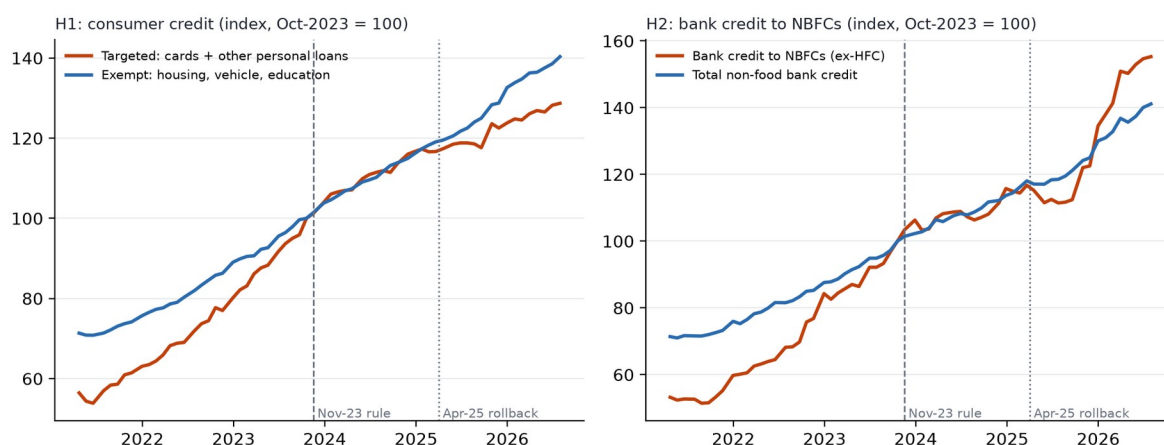
Table 6.3: Summary of hypotheses

Model	H0	H1 (alternative)	Test	Decision rule
1	$\delta_1 = 0$	$\delta_1 < 0$	t-test, Newey–West SE; panel with Driscoll–Kraay SE	Reject H0 if $p < 0.05$
2	$\delta_2 = 0$	$\delta_2 < 0$	t-test, Newey–West SE	Reject H0 if $p < 0.05$
3	$\beta_3 = 0$	$\beta_3 > 0$	Wild cluster bootstrap-t	Reject H0 if $p < 0.05$
4	$\beta_4 = 0$	$\beta_4 < 0$	Wild cluster bootstrap-t	Reject H0 if $p < 0.05$
5	$\beta_5 = 0$	$\beta_5 < 0$	t-test, Driscoll–Kraay SE	Reject H0 if $p < 0.05$

All *p*-values reported in this dissertation are two-sided, which is conservative relative to the one-sided alternatives.

7. EMPIRICAL FINDINGS AND RESULTS: H1 — CONSUMER-CREDIT CHANNEL

Figure 7.1: Bank credit indices, targeted versus exempt categories (October 2023 = 100)



Source: RBI DBIE Table 15; author's calculations. Indices chain monthly growth; July 2023 merger month set to zero growth for affected series.

Figure 7.1 (left panel) shows the raw data. Before November 2023, bank credit to cards and personal loans was growing much faster than exempt household credit and was catching up rapidly. After the policy, the two series moved together, and from 2025 targeted credit fell behind.

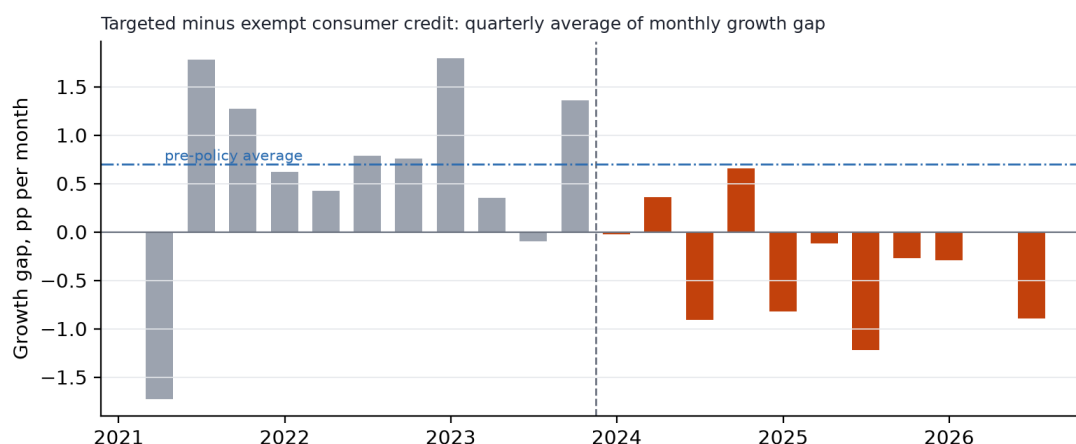
Table 7.1: H1 — effect of the consumer-credit risk weight on targeted credit growth

Specification	δ (pp per month)	Std. error	p-value	Annualised (pp)	N
Interrupted time series (Newey–West)	-0.66	0.22	0.002	-7.9	46
Panel DiD (Driscoll–Kraay)	-0.66	0.25	0.008	-7.9	229
Placebo Nov-2022, time series	0.45	0.33	0.175	5.4	31
Placebo Nov-2022, panel	0.46	0.38	0.231	5.5	154

Sample April 2021 – January 2025 (placebo: April 2021 – October 2023). δ is the change in the monthly growth gap between treated and control categories after the policy date.

The estimates support H1. After the policy, monthly growth of the targeted categories fell by 0.66 percentage points relative to exempt categories ($p = 0.002$); the panel specification gives the same point estimate with $p = 0.008$. Annualised, this is a slowdown of about 7.9 percentage points a year in targeted credit growth relative to exempt credit — large compared with the pre-policy growth rates of 20–23 per cent. The placebo estimates are positive and insignificant ($p = 0.175$ and 0.231): before the policy, if anything, the targeted categories were accelerating relative to the controls. The significance of the main estimate relies on autocorrelation-robust standard errors; Sections 7.3 and 8.5 report the classical ANOVA results and alternative inference methods, under which the evidence for H1 is weaker.

Figure 7.2: Monthly growth gap, targeted minus exempt consumer credit (quarterly averages)



Grey bars: pre-policy quarters; orange bars: post-policy quarters. Dashed line: pre-policy average gap.

7.1 Robustness

Table 7.2: H1 robustness (panel DiD, Driscoll–Kraay standard errors)

Specification	δ (pp per month)	Std. error	p-value	N
Drop credit cards	-0.79	0.19	< 0.001	183
Drop other personal loans	-0.53	0.38	0.164	183
Drop housing (merger-affected)	-0.71	0.26	0.006	184
Drop vehicle loans	-0.81	0.30	0.008	183
Drop education	-0.47	0.26	0.078	183
Add gold loans to control	-1.75	0.43	< 0.001	275
Shorter pre-period (from Apr-2022)	-0.78	0.21	< 0.001	169
Shorter post-period (to Oct-2024)	-0.67	0.27	0.013	214

The effect is negative in every specification. It is significant at the 5 per cent level in most, including when credit cards or the merger-affected housing series are dropped, when gold loans are added to the control group, and with shorter pre- or post-periods. Dropping other personal loans (leaving only credit cards as treated) or dropping education from the control group weakens significance to about the 10 per cent level, reflecting the smaller and more volatile series that remain.

7.2 Economic magnitude

To put the estimate in rupee terms, suppose targeted consumer credit had continued to grow at its pre-policy pace relative to exempt credit. Over the 15 months from November 2023 to January 2025, a monthly shortfall of 0.66 percentage points accumulates to about 10.4 per cent. With ₹18.0 lakh crore of bank credit outstanding to cards and other personal loans in January 2025, this implies roughly ₹1.9 lakh crore less targeted credit than would otherwise have been extended. This is an

approximate figure: it assumes the pre-policy relative growth would have continued, which may overstate the counterfactual given how fast unsecured credit had been growing.

7.3 Model summary, ANOVA and model equation

Table 7.3: Model 1 — model summary

Model	R	R ²	Adjusted R ²	Std. error of estimate	Durbin–Watson
1	0.20	0.04	0.02	1.52	2.21

Predictor: (Constant), Post. Dependent variable: Gap1 (targeted minus exempt consumer credit growth).

ANOVA. Null hypothesis: Post has no effect on the growth gap ($\delta = 0$). **Alternative hypothesis:** $\delta \neq 0$.

Table 7.4: Model 1 — ANOVA

	Sum of squares	df	Mean square	F	Sig.
Regression	4.27	1	4.27	1.85	0.181
Residual	101.52	44	2.31		
Total	105.78	45			

Classical OLS F-test, which assumes serially uncorrelated, homoskedastic errors.

Table 7.5: Model 1 — coefficients

	B	Std. error (Newey–West)	t	Sig.	VIF
(Constant)	0.67	0.17	3.96	< 0.001	
Post	-0.66	0.22	-3.07	0.002	1.00

Model equation: $Gap1(t) = 0.67 - 0.66 \cdot Post(t)$

Interpretation. Model 1 explains only 4.0 per cent of the month-to-month variation in the growth gap ($R^2 = 0.04$), and the classical F-test does not reject the null at conventional levels ($F = 1.85$, $p = 0.181$). This is expected for monthly growth data, which are dominated by short-term noise and seasonal swings that the policy dummy is not meant to explain; the relevant question is whether the average gap shifted, not whether the model predicts each month. The classical F-test also assumes serially independent errors, which the diagnostic tests in Section 8.4 do not fully support. With Newey–West standard errors the post-policy coefficient is significant ($t = -3.07$, $p = 0.002$), and the panel version with Driscoll–Kraay standard errors gives $p = 0.008$. Section 8.5 shows how much this conclusion depends on the choice of standard errors.

8. EMPIRICAL FINDINGS AND RESULTS: H2 — BANK-FUNDING CHANNEL AND FIRM-LEVEL EVIDENCE

8.1 Aggregate evidence

The right panel of Figure 7.1 compares bank credit to NBFCs (excluding HFCs) with total non-food bank credit. Before the policy, bank credit to NBFCs grew much faster than overall bank credit. After November 2023 it grew more slowly, and it accelerated sharply again after the April 2025 rollback.

Table 8.1: H2 — effect of the risk weight on bank exposures to NBFCs

Specification	δ (pp per month)	Std. error	p-value	Annualised (pp)	N
Main: NBFCs ex-HFC minus total non-food credit	-1.00	0.50	0.049	-11.9	45
Placebo Nov-2022	0.27	0.61	0.659	3.3	30
Alternative: NBFCs ex-HFC minus HFCs	-1.41	0.88	0.109	-16.9	45
Alternative: all NBFCs (incl. HFCs) minus non-food	-0.63	0.42	0.130	-7.6	45
Rollback: change after Apr-2025 (Dec-23 to Jul-26)	0.95	0.72	0.186	11.4	32

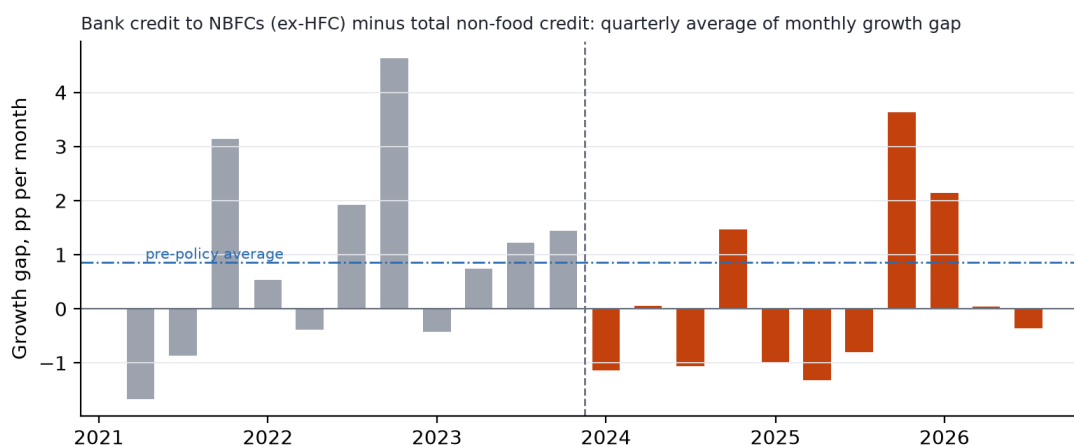
Interrupted time series on the monthly growth gap with Newey–West standard errors (6 lags). Main sample April 2021 – January 2025.

The main estimate supports H2: after the policy, monthly growth of bank credit to NBFCs (excluding HFCs) fell by 1.00 percentage points relative to total non-food credit ($p = 0.049$), about 11.9 percentage points a year. The placebo estimate is small, positive and insignificant ($p = 0.659$).

In rupee terms, a monthly shortfall of 1.00 percentage points over 15 months cumulates to about 16.1 per cent. With ₹12.9 lakh crore of bank credit to NBFCs (excluding HFCs) outstanding in January 2025, this implies roughly ₹2.1 lakh crore less bank funding to NBFCs than if the pre-policy relationship had continued — a substantial amount relative to the sector's funding needs, which NBFCs had to raise elsewhere or forgo.

Two alternative comparisons point in the same direction but are less precise. Against bank credit to the exempt HFCs, the slowdown is larger (16.9 percentage points a year) but noisier ($p = 0.109$), because the HFC series is small and was disrupted by the HDFC merger. Including HFCs in the treated group dilutes the effect ($p = 0.130$), as expected when exempt borrowers are mixed into the treated group. After the April 2025 rollback the gap widened again in favour of NBFCs by 0.95 percentage points a month ($p = 0.186$); the rebound is economically large but, with only 16 months of post-rollback data, not yet statistically significant. Bank credit to NBFCs is volatile from month to month because banks lend heavily at quarter-ends, which limits the precision of all H2 estimates.

Figure 8.1: Monthly growth gap, bank credit to NBFCs (ex-HFC) minus total non-food credit (quarterly averages)



Grey bars: pre-policy quarters; orange bars: post-policy quarters (including after the April 2025 rollback).

8.2 How did listed NBFCs adjust? Firm-level evidence

Company disclosures. Two lenders quantified the immediate capital impact of the rule in their December 2023 presentations, and one disclosed a pricing response (Table 8.2).

Table 8.2: Company statements on the effect of the November 2023 measures

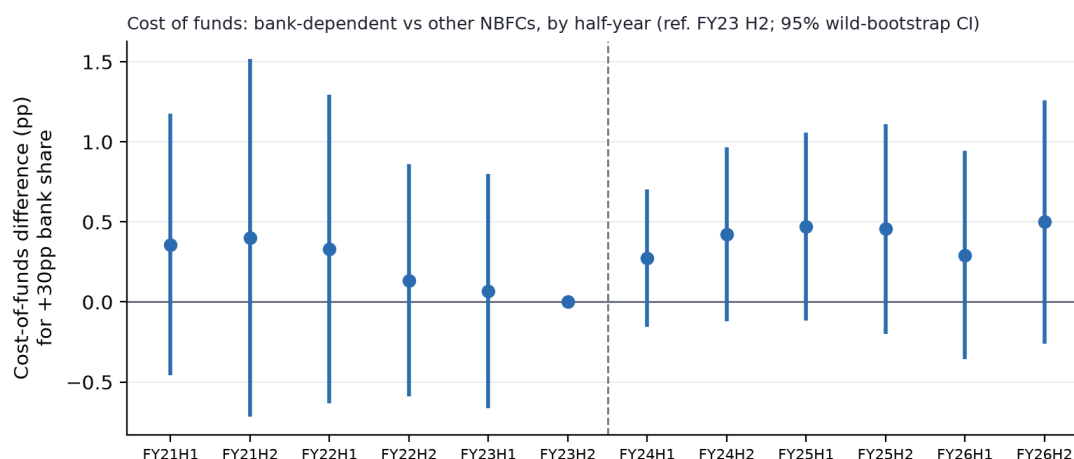
Company	Statement (verbatim)	Implication
Bajaj Finance (Q3 FY24)	“RBI increased risk weights on consumer credit exposure from 100% to 125% which had an impact of 290 bps on the Company's CRAR. Adjusted for this change CRAR would have been 26.77%.”	Capital ratio –2.9 pp
Bajaj Finance (Q3 FY24)	“Given the increase in risk weights and higher incremental cost of funds, the Company has increased rates across all portfolios by 20–30 bps.”	Lending rates +0.20–0.30 pp
SBI Cards (Q3 FY24)	“CRAR impacted by ~400 bps due to increase in risk weight by RBI.”	Capital ratio about –4 pp

Source: company investor presentations for the quarter ended 31 December 2023.

These statements confirm the mechanism behind H1: the higher risk weights immediately reduced the capital ratios of consumer lenders, and at least one passed part of the cost on to borrowers.

Cost of funds. Using 366 company-quarters for 16 firms, NBFCs that were more dependent on bank borrowing before the policy saw their cost of funds rise relative to others: the estimate implies about 0.23 percentage points higher cost of funds for a firm with 30 percentage points more bank funding. The effect builds up after the policy (Figure 8.2), but it is not statistically significant (wild-bootstrap $p = 0.283$), and part of it may reflect the slower repricing of bank loans after the 2022–23 rate rises.

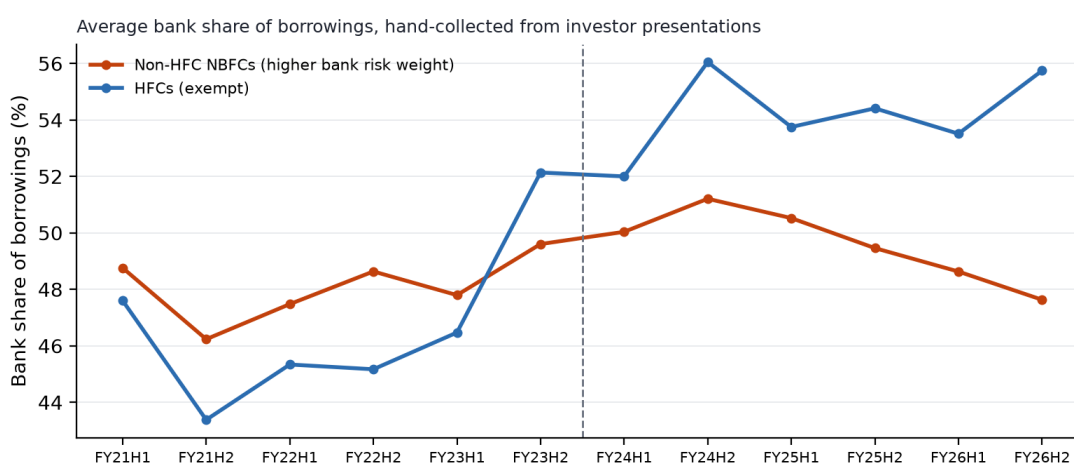
Figure 8.2: Cost of funds of bank-dependent versus other NBFCs, by half-year



Coefficients on bank share $\times$ half-year, scaled to a 30 pp difference in bank share; reference period FY23 H2; 95% wild-cluster-bootstrap intervals.

Bank share of borrowings. Exempt HFCs increased their reliance on bank borrowing after the policy, while non-HFC NBFCs did not (Figure 8.3). Relative to HFCs, the bank share of non-HFC NBFCs' borrowings fell by 3.3 percentage points before the rollback ($p = 0.280$) and by about 6.1 percentage points by the second half of FY2026 ($p = 0.119$). The placebo is insignificant ($p = 0.565$). The direction is consistent with the aggregate H2 result — banks lent less to NBFCs, so NBFCs relied more on bonds, commercial paper and external borrowing — but with 15 firms the firm-level difference is not statistically significant.

Figure 8.3: Average bank share of borrowings, non-HFC NBFCs versus HFCs



Hand-collected from borrowing-mix slides of quarterly investor presentations; definitions follow each company's disclosure.

Within-firm lending: Bajaj Finance. Bajaj Finance is the largest NBFC consumer lender in the sample and reports assets by business segment every quarter. Its consumer segments (sales finance and B2C personal loans) made up 37.1 per cent of assets in September 2023, dipped to 35.7 per

cent by March 2024 and recovered to 37.1 per cent by March 2025. Comparing the growth of affected and exempt segments within the firm, there is no significant relative slowdown (0.58 pp per quarter, $p = 0.749$). Bajaj absorbed the capital impact, raised its rates and continued to grow its consumer book.

Table 8.3: Firm-level tests (hand-collected data)

Test	Estimate	p-value	Obs.	Firms
Cost of funds $\sim$ Post $\times$ pre-policy bank share	0.23 pp per 30 pp share	0.283	366	16
Bank share $\sim$ Post $\times$ non-HFC (to Mar-2025)	-3.29 pp	0.280	235	15
Bank share, placebo Nov-2022	-1.62 pp	0.565	—	15
Bajaj: affected vs exempt segment growth	0.58 pp per quarter	0.749	171	9 segments

Two-way fixed effects (firm/segment and quarter). Firm tests: wild cluster bootstrap-t by firm. Bajaj: Driscoll–Kraay standard errors.

8.3 Model summary, ANOVA and model equation for Model 2

Table 8.4: Model 2 — model summary

Model	R	R ²	Adjusted R ²	Std. error of estimate	Durbin–Watson
2	0.20	0.04	0.02	2.27	2.01

Predictor: (Constant), Post. Dependent variable: Gap2 (NBFC ex-HFC minus non-food credit growth).

ANOVA. Null hypothesis: Post has no effect on the growth gap ($\delta = 0$). **Alternative hypothesis:** $\delta \neq 0$.

Table 8.5: Model 2 — ANOVA

	Sum of squares	df	Mean square	F	Sig.
Regression	9.55	1	9.55	1.86	0.180
Residual	220.89	43	5.14		
Total	230.45	44			

Classical OLS F-test, which assumes serially uncorrelated, homoskedastic errors.

Table 8.6: Model 2 — coefficients

	B	Std. error (Newey–West)	t	Sig.	VIF
(Constant)	0.89	0.43	2.08	0.037	
Post	-1.00	0.50	-1.97	0.049	1.00

Model equation: $Gap2(t) = 0.89 - 1.00 \cdot Post(t)$

Interpretation. As with Model 1, the policy dummy explains only a small share of monthly variation ($R^2 = 0.04$) and the classical F-test is not significant ($F = 1.86$, $p = 0.180$). Bank lending to NBFCs surges at quarter-ends and falls back in the following month, so much of the residual variation is seasonal. With Newey–West standard errors the post-policy coefficient is significant at

the 5 per cent level ($p = 0.049$), and Section 8.5 shows that the result is significant under several methods once calendar-month seasonality is controlled for.

8.4 Diagnostic tests

Table 8.7: Diagnostic tests on the residuals of Models 1 and 2

Test	Null hypothesis	Model 1 statistic (p)	Model 2 statistic (p)
Jarque–Bera (normality)	Residuals are normally distributed	2.27 (0.322)	3.41 (0.181)
Skewness / kurtosis	—	0.44 / 3.64	0.66 / 3.27
Durbin–Watson	No first-order autocorrelation (≈ 2)	2.21	2.01
Ljung–Box Q (12 lags)	No autocorrelation up to lag 12	19.61 (0.075)	8.18 (0.771)
Breusch–Pagan	Homoskedastic residuals	3.15 (0.076)	1.47 (0.226)
VIF (Post)	No multicollinearity ($VIF < 10$)	1.00	1.00
VIF (Treated $\times$ Post, H1 panel with fixed effects)	No multicollinearity ($VIF < 10$)	2.10	—

Residuals are consistent with normality in both models (Jarque–Bera $p = 0.322$ and 0.181), and multicollinearity is not a concern (all VIFs well below 10; a single regressor has $VIF = 1$). The Durbin–Watson statistics are close to 2, indicating no first-order autocorrelation. For Model 1, however, the Ljung–Box test points to autocorrelation at longer lags ($p = 0.075$, driven by a negative correlation at lag 2) and the Breusch–Pagan test to some heteroskedasticity ($p = 0.076$), both significant at the 10 per cent level. Model 2 shows neither. These results justify autocorrelation- and heteroskedasticity-robust standard errors for Model 1, but they also mean that Model 1's significance depends on how that correction is made.

8.5 Sensitivity of inference to the choice of standard errors

Because each main test uses only 45–46 monthly observations, it is important to check whether the conclusions depend on the method used to compute standard errors. Table 8.8 re-estimates the post-policy shift for both models under eleven methods: classical OLS; White heteroskedasticity-robust errors; Newey–West errors with 1, 3 (the common rule of thumb for this sample size), 6 (the main specification) and 12 lags; the same model with calendar-month dummies to remove seasonality; and a moving-block bootstrap (blocks of three months, 10,000 replications) that does not rely on any asymptotic formula.

Table 8.8: Post-policy coefficient and p-value under alternative inference methods

Method	Model 1 δ	Model 1 p	Model 2 δ	Model 2 p
Classical OLS	-0.66	0.181	-0.99	0.180
White (HC1)	-0.66	0.089	-0.99	0.122
Newey–West, 1 lag	-0.66	0.076	-0.99	0.104

Method	Model 1 δ	Model 1 p	Model 2 δ	Model 2 p
Newey-West, 3 lags (rule of thumb)	-0.66	0.010	-0.99	0.086
Newey-West, 6 lags (main)	-0.66	0.002	-0.99	0.049
Newey-West, 12 lags	-0.66	< 0.001	-0.99	0.026
Month dummies, classical OLS	-0.69	0.107	-1.18	0.065
Month dummies, White (HC1)	-0.69	0.102	-1.18	0.043
Month dummies, Newey-West 3 lags	-0.69	0.007	-1.18	0.012
Moving-block bootstrap (block 3)	-0.66	0.106	-0.99	0.180
Month dummies, moving-block bootstrap	-0.69	0.064	-1.18	0.047

Sample April 2021 – January 2025. Month dummies: eleven calendar-month indicators added to the regression. Block bootstrap: residuals resampled in blocks of three months under the null of no effect.

Three conclusions follow. First, the size and sign of the estimates are stable: every method gives a negative post-policy shift of about 0.66–0.69 percentage points a month for Model 1 and 0.99–1.18 for Model 2. Second, Model 2 is robust once seasonality is removed: with calendar-month dummies it is significant at the 5 per cent level with White errors ($p = 0.043$), Newey–West errors ($p = 0.012$) and the block bootstrap ($p = 0.047$). Third, Model 1 is less robust: it is significant at the 1–5 per cent level with Newey–West errors of three or more lags (with or without seasonal dummies) and in the panel specification, but only at about the 6–11 per cent level with White errors or the block bootstrap, and not significant with classical OLS errors ($p = 0.181$).

H2 is therefore supported with reasonable confidence. H1 is supported in the main specifications and is consistent in direction, magnitude, placebo tests and company disclosures, but its statistical significance should be described as moderate: it depends on correcting for the serial correlation in the monthly data, and a cautious reader could regard it as significant only at the 10 per cent level.

9. DISCUSSION

9.1 Interpretation

Both hypotheses are supported at the system level, H2 with reasonable confidence and H1 with moderate confidence (its significance depends on correcting for serial correlation; Section 8.5). Raising the capital requirement on unsecured consumer credit slowed bank credit to cards and personal loans by about eight percentage points a year relative to exempt household credit. Raising the capital requirement on bank loans to NBFCs slowed bank lending to NBFCs by about twelve percentage points a year relative to total bank credit, and the partial reversal in April 2025 was followed by a rapid recovery in NBFC funding while card credit — still subject to the higher risk weight — stayed weak. These patterns are what a targeted capital-requirement shock would produce, and they are not visible before the policy.

At the level of listed NBFCs, the adjustment looks different. The rule cut capital ratios immediately (by about 3 to 4 percentage points at the two lenders that disclosed it), and at least one lender raised its lending rates. Bank-dependent NBFCs saw somewhat higher funding costs, and non-HFC NBFCs gradually reduced their reliance on bank borrowing. But large listed NBFCs did not shrink their consumer lending relative to their other lending. These firms are well capitalised, rated AAA or AA, and can raise equity and bond finance: Bajaj Finance and Cholamandalam both raised fresh equity around the time of the policy. The policy therefore appears to have bitten mainly through banks' own lending decisions, while large NBFCs absorbed the shock through capital, pricing and funding substitution.

This answers the question posed in the synopsis — whether NBFCs reduced overall lending or adjusted in other ways. For the large listed NBFCs studied here, the answer is the latter. The very rapid growth of exempt, secured gold loans in the banking data after the policy is also consistent with lenders shifting toward secured credit.

9.2 Relation to the synopsis hypotheses

The synopsis proposed five hypotheses. Following the guide's advice the formal tests focus on two, but the evidence speaks to all five:

- **H1 (unsecured exposure and loan growth):** supported at the system level — bank credit to the targeted unsecured categories slowed significantly. Within Bajaj Finance, the largest NBFC consumer lender, consumer segments did not slow relative to other segments.

- **H2 (bank dependence and borrowing costs):** the banking system cut its lending to NBFCs significantly. Firm-level funding costs of bank-dependent NBFCs rose somewhat more than others, but not significantly; the 2022–23 rate cycle dominated funding costs.
- **H3 (capital):** company disclosures show the policy reduced capital ratios of consumer lenders by about 3–4 percentage points; the largest affected lenders responded by raising equity, so the slowdown did not need to come from their lending.
- **H4 (shift to secured lending):** in the banking data, exempt and secured gold loans grew by 62.4 per cent a year after the policy, far faster than before (6.8 per cent), consistent with a shift toward secured credit.
- **H5 (recovery after February 2025):** bank credit to NBFCs recovered quickly in aggregate after the April 2025 rollback (21.4 per cent a year), but at the firm level non-HFC NBFCs continued to reduce their reliance on bank borrowing, consistent with the synopsis expectation that the relaxation would not bring an immediate full return to bank funding.

9.3 Policy implications

- Targeted risk weights are an effective tool for slowing specific segments of bank credit without a broad tightening: exempt categories continued to grow.
- The bank-to-NBFC lever transmits quickly and reverses quickly: bank credit to NBFCs slowed after November 2023 and recovered after April 2025.
- For NBFC managers, diversified funding (bonds, external commercial borrowing, securitisation) reduced exposure to the bank-funding channel; firms with concentrated bank funding were most exposed to this kind of regulatory change.
- Well-capitalised NBFCs can offset part of the intended effect through capital raising and alternative funding; regulators assessing the policy should look at NBFC balance sheets as well as bank credit data.

9.4 Limitations

- The RBI data measure bank credit, not NBFCs' own lending; the aggregate tests capture the banking system's response.
- The difference-in-differences design assumes that, without the policy, the growth gap between treated and control categories would have remained at its pre-policy average. The placebo tests support this, but it cannot be proven.
- Bank credit to NBFCs is volatile from month to month and seasonal (quarter-end lending); alternative comparison groups for H2 give p-values of about 0.11–0.13.

- With 45–46 monthly observations, the significance of the aggregate tests depends on the standard-error method. The classical OLS F-test is not significant for either model ($p \approx 0.18$); H2 is significant under several methods once seasonality is controlled for, while H1 is significant with Newey–West and Driscoll–Kraay standard errors but only at about the 10 per cent level with White errors or a block bootstrap (Table 8.8).
- The firm-level sample has only 16 firms, which limits statistical power; firm definitions of cost of funds and bank borrowing differ, which is why all firm-level comparisons are within-firm over time.
- Some firm-level series have breaks: Shriram Finance is included from FY2023 (after the merger of Shriram Transport Finance and Shriram City Union Finance); Poonawalla Fincorp from FY2022 (after the change of control of Magma Fincorp); Arman Financial's early borrowing figures are partly interpolated.
- Following the guide's advice, the study focuses on two hypotheses. Synopsis hypotheses on capital buffers (H3), secured-loan substitution (H4) and the speed of recovery after the 2025 relaxation (H5) are addressed only descriptively.

10. CONCLUSION

This dissertation evaluated the RBI's November 2023 increase in risk weights on consumer credit and on bank lending to NBFCs. Using RBI sectoral credit data and a difference-in-differences design, it finds that both measures reduced the targeted flows of bank credit: targeted consumer credit grew about 7.9 percentage points a year more slowly than exempt household credit ($p = 0.002$), and bank credit to NBFCs grew about 11.9 percentage points a year more slowly than total bank credit ($p = 0.049$). Placebo tests show no such effects before the policy. The bank-funding result is robust across inference methods once seasonality is controlled for; the consumer-credit result is consistent in direction and size but statistically more fragile, being significant with autocorrelation-robust standard errors and at about the 10 per cent level under more conservative methods. A firm-level dataset hand-collected from 382 investor presentations shows how listed NBFCs absorbed the shock — through lower capital ratios, higher lending rates, somewhat higher funding costs and a gradual shift away from bank borrowing — rather than by cutting their own lending. The results suggest that targeted capital requirements are a precise and reversible tool for managing credit risk in specific segments, and that their effect on the NBFC sector depends on how easily NBFCs can replace bank funding.

Future work could extend the firm-level dataset to unlisted and smaller NBFCs, which are more likely to be funding-constrained, and use loan-level or bureau data to measure NBFCs' consumer lending directly.

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Company investor presentations, Q1 FY2021 – Q4 FY2026: AAVAS Financiers, Arman Financial Services, Bajaj Finance, Can Fin Homes, Chola mandalam Investment & Finance, CreditAccess Grameen, Home First Finance, IIFL Finance, LIC Housing Finance, Mahindra & Mahindra Financial Services, Muthoot Finance, PNB Housing Finance, Poonawalla Fincorp, Repco Home Finance, SBI Cards and Payment Services, Shriram Finance.

APPENDICES

Appendix A: Data Files and Reproducibility

All data used in this dissertation are provided in the accompanying workbook `Dissertation_Data.xlsx`, which contains the following sheets:

- `README` — definitions and sources of all variables.
- `RBI_Monthly` — outstanding bank credit by category and monthly growth rates.
- `RBI_Growth_Table` — the figures in Table 4.1.
- `COF_Quarterly` and `BankShare_Quarterly` — hand-collected firm data, with the presentation and page each value came from.
- `Bank_Exposure_Sep2023` — verified pre-policy bank shares and the values in the original dataset.
- `Bajaj_Segments` — segment-wise assets of Bajaj Finance.
- `Company_Disclosures` — verbatim company statements in Table 8.2.
- `Results_*` — all regression results reported in Chapters 7 and 8.

The analysis was carried out in Python (`pandas`, `statsmodels` and `linearmodels`). The scripts are `05_final_analysis.py` (all results, tables and figures) and `did_tools.py` (two-way fixed-effects estimation with wild cluster bootstrap-t inference).

Appendix B: Firm-Level Data Notes

- Cost of funds is the company's reported cost of borrowing where available. For Mahindra & Mahindra Financial Services, Shriram Finance and Arman Financial Services it is computed as annualised quarterly finance cost divided by average borrowings from the same presentation; for Cholamandalam it is finance cost as a share of average assets (the only consistently reported measure).
- Where a later presentation restates an earlier quarter, the value as first reported is used.
- Muthoot Finance's FY2021–22 figures are derived from reported year-to-date averages.
- Bank share of borrowings follows each company's own borrowing-mix definition (for example, banks and financial institutions for Mahindra & Mahindra; term loans for Shriram and AAVAS). Cholamandalam's institutional financial investors (IFI), reported separately from FY2024, are added back to banks for consistency.
- IIFL Finance's bank share is available only annually on a consistent definition; PNB Housing's earlier figures are rescaled to exclude direct assignment.

Appendix C: Data Tables

Table C.1: Cost of funds by company, annual average of quarterly values (per cent)

Company	FY21	FY22	FY23	FY24	FY25	FY26
AAVAS Financiers	7.77	7.08	7.19	7.88	8.18	7.79
Arman Fin. Services	10.96	10.60	12.16	12.20	11.94	11.49
Bajaj Finance	7.39	6.83	7.02	7.72	7.96	7.54
Can Fin Homes	7.00	5.59	6.25	7.41	7.55	7.21
Cholamandalam Inv. & Fin.	6.70	5.82	6.12	6.93	7.07	6.78
CreditAccess Grameen	9.32	9.35	9.32	9.75	9.80	9.48
Home First Finance	8.05	7.18	7.32	8.15	8.38	8.12
IIFL Finance	9.05	8.72	8.75	9.07	9.22	9.39
LIC Housing Finance	7.42	6.71	7.21	7.68	7.75	7.37
Mahindra & Mahindra Fin. Services	7.82	6.93	6.91	7.65	7.74	7.46
Muthoot Finance	9.10	8.30	8.23	8.78	8.92	8.78
PNB Housing Finance	7.86	7.28	7.46	8.00	7.86	7.58
Poonawalla Fincorp	—	8.23	7.39	8.04	8.10	7.75
Repcos Home Finance	7.92	7.02	7.42	8.32	8.80	8.52
SBI Cards and Payment Services	6.20	5.25	5.88	7.30	7.38	6.60
Shriram Finance	—	—	8.20	8.85	8.93	8.94

Financial years ending March. Source: hand-collected from quarterly investor presentations (see Appendix B for definitions).

Table C.2: Bank share of borrowings by company, annual average of quarterly values (per cent)

Company	FY21	FY22	FY23	FY24	FY25	FY26
AAVAS Financiers	35.2	37.3	43.4	48.6	50.8	—
Arman Fin. Services	29.0	29.1	28.3	35.8	35.6	27.2
Bajaj Finance	35.8	29.2	31.5	31.5	29.0	27.8
Can Fin Homes	55.5	49.0	54.0	58.0	56.0	60.0
Cholamandalam Inv. & Fin.	66.0	66.2	65.2	60.0	58.5	58.2
CreditAccess Grameen	53.6	53.6	59.1	54.1	58.5	57.0
Home First Finance	44.0	45.0	53.2	56.0	59.5	59.0
IIFL Finance	28.0	35.0	45.0	51.0	35.0	42.0
LIC Housing Finance	23.1	27.6	33.7	33.2	33.2	35.8
Mahindra & Mahindra Fin. Services	50.5	47.0	54.8	57.6	60.6	64.1
Muthoot Finance	41.8	49.6	57.3	62.4	55.2	49.7
PNB Housing Finance	26.7	32.2	34.1	40.2	38.7	37.0
Repcos Home Finance	72.0	68.5	72.2	76.5	81.6	84.1
SBI Cards and Payment Services	58.5	60.5	68.2	78.5	83.8	84.8
Shriram Finance	—	—	26.3	25.1	22.4	19.7

Financial years ending March. Poonawalla Fincorp does not disclose its borrowing mix. Source: hand-collected from quarterly investor presentations.

Table C.3: Outstanding bank credit by category at end-March (₹ lakh crore)

Month	Cards	Oth. PL	Housing	Vehicle	Educ.	NBFC ex-HFC	HFCs	Non-food
Mar-2019	0.88	5.53	11.77	2.85	0.76	5.26	2.26	97.30
Mar-2020	1.20	6.94	13.62	3.40	0.78	7.01	2.49	103.19
Mar-2021	1.32	7.63	14.92	3.68	0.78	6.64	2.84	108.88
Mar-2022	1.55	9.02	17.38	4.05	0.84	7.47	2.86	118.36
Mar-2023	2.05	11.62	19.91	4.88	0.96	10.24	3.19	136.55
Mar-2024	2.57	14.28	27.19	5.73	1.19	12.02	3.21	165.64
Mar-2025	2.84	15.35	30.10	6.23	1.37	13.12	3.23	183.72
Mar-2026	2.94	17.32	33.56	7.39	1.56	16.96	3.73	212.91

Source: RBI DBIE Table 15. Figures from July 2023 include the effect of the HDFC merger (housing up, HFC and NBFC credit down).